

Question 1

- For (C) , instead of separate 14 files, we can make a pipeline with dvc. We can just run one command-dvc repro. Training stage with MLflow and commit dvc.yaml to git

Question 2

Across the six experimental runs, Run 4 (learning_rate=0.001, hidden_layer_sizes=(128, 64)) achieved the best overall performance, attaining the highest validation accuracy of 95.40% (0.954) and the lowest training loss of 0.003. The deeper two-layer architecture effectively captured the complex spatial representations of handwritten digits without destabilizing.

Overfitting vs. Optimization Divergence:

Rather than standard overfitting (where training loss continues to drop while validation accuracy degrades), aggressive learning rates (learning_rate=0.1) caused severe optimization divergence:

- In the single-layer setup (64,) at $\text{lr}=0.1$, training loss remained elevated at 0.534 with validation accuracy dropping to 79.00%.
- In the deeper setup (128, 64) at $\text{lr}=0.1$, training loss spiked to 1.091 and validation accuracy degraded sharply to 62.80%.

At the optimal learning rate of 0.001, both architectures converged smoothly without evidence of overfitting within the 20-epoch constraint.

Hyperparameter Sensitivity:

The learning rate had a significantly larger impact on model performance than network depth. Expanding hidden layers from (64,) to (128, 64) yielded an accuracy improvement of 1.4% at $\text{lr}=0.001$ (from 0.940 to 0.954). In contrast, scaling the learning rate from 0.001 to 0.1 degraded performance by 15.0% in the single-layer architecture and 32.6% in the deeper network.

